



Dilruk Migara Wickramarachchi

Undergraduate - SLIIT

✉ dilrukmigara@gmail.com ☎ +94 77 97 80 053 📍 Malabe 🌐 migarawickramarachchi

🔗 dilrukmigara.me

Motivated and detail-oriented **Computer Systems Engineering undergraduate at SLIIT**, with a strong foundation in **electronics, programming, and control systems**. Skilled in applying engineering principles to real-world problem-solving, with hands-on experience in system design, simulation, and project work. Proficient in teamwork and communication, with a proven ability to balance technical expertise and leadership responsibilities. Actively developing knowledge in **embedded systems, circuit design, and software development**, while seeking opportunities to apply academic learning in a practical, industry-focused environment.

🎓 Education

05/2023 – Present	BSC (hon's) Computer Systems Engineering, <i>Sri Lanaka Institute Of Information Technology</i>
2020	Advance Level - Physical Science Stream, <i>C.W.W. KANNAGARA CENTRAL COLLAGE</i>

🧠 Technical Skills

- Programming: C, C++, Python, SQL, R-studio (basic), Acembly
- Electronics & Hardware: Circuit & PCB design, Arduino, Microcontrollers
- Control Systems: PID controllers, System Modeling, MATLAB/Simulink, Proteus
- Software Tools: MATLAB, Multisim, KiCAD , Logisim Git/GitHub
- Networking & Systems: Computer Networks, Operating Systems (Linux), Database Management, Cisco Networking (Basic) , Ubuntu Linux (Familiar), Kali Linux (Familiar), Information Security Fundamentals (Currently Learning)
- Mathematics & Engineering Knowledge: Engineering Mathematics, Electronic Theory & Circuit Analysis, Strong foundation in analytical and problem-solving approaches for engineering applications
- Foundational knowledge in Machine Learning and Image Processing gained through structured online courses
- Artificial Intelligence: Foundational knowledge in Machine Learning and Image Processing, Deep Learning Fundamentals (Currently Learning), Computer Vision, TinyML

👤 Professional Skills

- Leadership & Team Management
- Problem Solving & Analytical Thinking
- Communication & Collaboration
- Marketing Strategy & Event Promotion (Goal Marketing – Startup Experience)
- OL maths tuition master (mathsbook.dilrukmigara.me)

IoT Pill Reminder System – Ideal Spark Exhibition (SLIIT, 2024)

- Designed and developed a complete **IoT-based pill reminder system** using Arduino and sensors, focused on healthcare and meditation support.
- Showcased at the *Ideal Spark Exhibition* and later enhanced with improved features for real-world application.

ERP Web Application Development (2024)

- Built a **web-based ERP system** using **Java, Spring Boot, MySQL, and React**.
- Implemented REST APIs, user authentication, and database integration for enterprise resource management.

Analog Electronics Project – Multi-Stage Amplifier & Filtering Circuit (2024)

- Designed and implemented a **BJT-based amplifier** with **high-pass and low-pass filters** for AM radio audio signal processing.
- Tested performance and ensured clear audio output through signal filtering.

Multicore Computer System Simulation (2024)

- Designed and evaluated a simulated **4-core processor system** to analyze **ILP, multithreading, and multicore architectures**.
- Measured performance metrics including **execution time, CPU utilization, and context switching** under varying workloads.

Community Radio FM Transmitter Project (2024)

- Designed and built a **low-power FM transmitter** using **modulation ICs, antenna design, and audio input integration**.
- Successfully broadcasted within the FM band (88–108 MHz), compatible with standard FM receivers.

DC Motor Position Control Using PID

- Designed and simulated a **PID controller** to rotate a DC motor from 0° to 90° with **minimal overshoot and fast settling time**.
- Implemented the controller in **MATLAB/Simulink**, analyzing system response, stability, and disturbance rejection.
- Demonstrated practical application of **control theory** in automated positioning systems.

PCB Design & Fabrication Project

- Designed and simulated a **custom PCB circuit** using **KiCad and Proteus** for schematic capture and layout verification.
- Fabricated and **hand-assembled the PCB**, testing its functionality and ensuring correct operation.
- Gained practical experience in **circuit design, PCB layout, and hardware prototyping**.

Maze-Solving Robot – SLIIT (2025)

- Designed and developed an autonomous maze-solving robot using Arduino Uno, IR sensors, and DC motors.
- Implemented an optimized maze-solving algorithm with modular C programming for reliable navigation and performance testing.

Animal Detection & Smart Farm Monitoring System (2024)

- Developed an **AI-powered smart farm system** using **ESP32 Dev Board and ESP32-CAM** with IoT integration and a Flask-based web server for real-time monitoring.
- Implemented a **YOLOv8 deep learning model** for real-time animal detection, tracking, and alert generation from live camera streams with sensor-based environmental monitoring.

 References

Prof. Anuradha Jayakody, Head of the Department of Computer Systems Engineering,
SLIIT Faculty of Computing
anuradha.j@sliit.lk, 0117543923

★ Extracurricular activities

2025 – 2026	Faculty of computing student Community(FCSC) - SLIIT • Position - commity member
2025 – 2026	Computer Systems Engineering Student Community- SLIIT • Position - vice President
2015 – 2020	National Cadet Corps • Company Sargent Major (CSM) in 12th battalion - Kaluthara • Platoon Sergeant in school. • Advance course and sargent course done at rantabe • All 6 camps attented in rantabe.
2020 – 2024	National youth crop • Vise president - distric federation of youth club Kaluthara • President - devisional federation of youth club , Palindanuwara • Attented youth leadership camps in maharagama youth center • Attended to national events.
2011 – 2020	School achivements • School brass band leader • Senior prefect in school • School house meet thissa house vice captan • Cadet sargent • Karate player • Member school miusic band
2014	Sport Achievements • Western province karate meet 2016 • Shotokan karate meet 2016 • School inter house meet karate • School inter house meet Javelin • Mathugama educational zone Javelin • School inter house meet cricket team & elle team • Youth corps , divisional meet Javelin & kabadi & cricket

Summary

Dedicated and disciplined leader with a strong background in team management, leadership, and sports. Former **Company Sergeant Major** of the National Cadet Corps with extensive youth leadership experience as **Vice President of the District Federation Youth Club** and **Senior Student Leader**. Currently serving as the **Vice President of the Computer Systems Engineering Student Committee** at SLIIT and an **active member of academic communities** contributing to knowledge sharing and student development. Also a committee member of the **Faculty of Computing Student Community (FCSC)**, supporting student engagement and event organization. Excels in both academics and athletics, with notable achievements in **martial arts** and **track and field**. Committed to personal growth, teamwork, and making a positive impact in every environment.

✂ Social References

Mr. Indika Sumanarathna

vice Principal ,C.W.W. Kannangara Centrala Collage ,Mathugama.
0775425177